

Multiple Choice

1. **Answer:** A

Options: A) $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$; B) $\text{PbH}_4 < \text{SnH}_4 < \text{CH}_4 < \text{GeH}_4 < \text{SiH}_4$; C) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$; D) $\text{CH}_4 < \text{PbH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{SiH}_4$

Note: The thermal stability of group-14 hydrides increases with decreasing atomic size and increasing bond strength; thus, CH_4 is the most stable due to the strong C-H bonds, while PbH_4 is the least stable due to the weak Pb-H bonds, resulting in the order $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$.

2. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: The Larmor frequency for a proton in a magnetic field is calculated using the formula $f = B$, where γ (the gyromagnetic ratio for a proton) is approximately 42.58 MHz/T, leading to a frequency of 42.58 MHz in a 1 T magnetic field.

3. **Answer:** A

Options: A) MgO; B) MgS; C) NaF; D) NaCl

Note: MgO has the greatest lattice enthalpy due to the high charge and small size of the magnesium and oxide ions, which results in stronger electrostatic forces of attraction between them compared to the other ionic substances.

4. **Answer:** A

Options: A) It has little effect.; B) It significantly increases the pH.; C) It significantly decreases the pH.; D) It changes the pH asymptotically to the pK_a of the acid.

Note: Doubling the volume of the buffer solution by adding water dilutes both the weak acid and its conjugate base equally, maintaining their ratio and thus having little effect on the pH according to the Henderson-Hasselbalch equation.

5. **Answer:** B

Options: A) NH_3 ; B) NH_2^- ; C) NH_4^+ ; D) N_2H_4

Note: NH_2^- is the strongest base in liquid ammonia because it is the conjugate base of ammonia (NH_3) and possesses a high affinity for protons due to its negative charge, making it more capable of accepting protons than other species present in that solvent.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because 2-methylpentane has more than three distinct

chemical shifts in its ^{13}C NMR spectrum due to the presence of multiple unique carbon environments resulting from its branching and chain structure.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because barium hydroxide cannot be directly converted to barium metal due to the need for a reduction process, which typically requires an intermediate step involving the removal of hydroxide ions or the presence of a reducing agent.

8. **Answer:** False

Options: A) True; B) False

Note: The equilibrium polarization of ^{13}C nuclei is higher at higher magnetic field strengths, such as 10.0 T, because the polarization increases with the strength of the magnetic field due to the greater difference in energy between the nuclear spin states.

9. **Answer:** True

Options: A) True; B) False

Note: The atomic radii of the lanthanide elements decrease from La to Lu because the increasing effective nuclear charge, resulting from the greater number of protons, pulls the electron cloud closer to the nucleus, effectively reducing the size of the atom.

10. **Answer:** True

Options: A) True; B) False

Note: In the balanced redox reaction between permanganate ion (MnO_4^-) and iodide ion (I^-), the stoichiometry shows that for every 6 moles of MnO_4^- reduced, 5 moles of I^- are oxidized, establishing the 6:5 ratio.

Long Form Response

11. **Answer:** The chemical environment of CH_3Cl significantly influences its proton NMR resonance frequency due to the presence of the electronegative chlorine atom, which creates a deshielding effect on the protons of the methyl group. In a 12.0 T magnetic field, this deshielding results in a downfield shift of the NMR signal, meaning the protons resonate at a higher frequency compared to protons in less electronegative environments. This observation is significant in spectroscopy as it illustrates how molecular structure and electronegativity affect the electronic environment of nuclei, thereby allowing chemists to deduce structural information about the compound. Understanding these shifts aids in the identification and characterization of organic molecules through NMR spectroscopy.

Note: correct because it accurately describes how the electronegative chlorine atom in CH_3Cl creates a deshielding effect on the protons, leading to a downfield shift in their NMR resonance frequency in a strong magnetic field, which is a key concept in understanding molecular environments in spectroscopy.

12. **Answer:** The ratio of line intensities in the EPR spectrum of the t-Bu radical ($(\text{CH}_3)_3\text{C}\cdot$) is influenced by several factors, including the multiplicity of the radical and the coupling of unpaired electrons with surrounding nuclei. The intensity distribution observed is a result of the relative populations of different spin states, which is reflected in the ratio of 1:8:28:56:70:56:28:8:1. This specific ratio arises from the combinatorial possibilities of arranging the unpaired electron spins among the equivalent hydrogen nuclei surrounding the carbon atom. As a result, the highest intensity corresponds to the most probable configuration, while the lower intensities represent less likely arrangements, leading to the characteristic pattern seen in the EPR spectrum.

Note: correct because it identifies that the ratio of line intensities in the EPR spectrum of the t-Bu radical is determined by the multiplicity of the radical and the arrangements of unpaired electron spins, leading to specific intensity distributions that reflect the combinations of different spin states.

End of Answer Key